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A Feasibility Study for Detecting Schwarzschild Apsidal Precession in the Orbit of S301 around Sgr A* with GRAVITY+/VLTI

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ABSTRACT

S301 is a recently discovered star (GRAVITY Collaboration et al. 2026). It orbits Sgr A* on a highly eccentric orbit ($e = 0.9832$) with a period of 8.68 yr. At periapsis, S301 reaches 136 Schwarzschild radii from Sgr A* and exceeds 8% of the speed of light. S301's orbit should show rosette-shaped apsidal precession, driven by ordinary first-post-Newtonian (1PN) general relativity. This precession is the same effect that shifts Mercury's perihelion, scaled up here by S301's proximity to a far more compact central body. The precession does not depend on Sgr A*'s spin and does not require a spin measurement first. Sgr A*'s spin was the original motivation for discovering S301. This paper asks whether a real, near-term, ground-based campaign can detect S301's Schwarzschild precession from astrometry alone. We build the analysis in the order a real observing proposal would need: first validate the physics model, then design the observing campaign, then fit the orbit from the recovered data. Concretely, we validate a relativistic light-curve model against the published orbital solution, design a synthetic two-passage GRAVITY+ campaign, and fit a seven-parameter orbit to the simulated astrometry. Two of our own initial design choices fail real tests along the way. The first choice is a proposed radial-velocity channel on ERIS (the Enhanced Resolution Imager and Spectrograph). Radial velocity is the usual way S-star orbits are solved, so we considered adding it. At S301's faint magnitude, this radial-velocity channel would add no measurable precision to the fit. We confirmed this directly, rather than assuming it, so radial velocity was never proposed for the actual campaign. The second choice is an observing cadence chosen by Fisher-information (D-optimal) design. We expected this design to beat a simple heuristic cadence. Instead, the Fisher-optimal cadence collapses when tested in a real nonlinear orbit fit, despite an excellent analytic prediction. This failure is a reproducible pitfall: local-linearization design breaks down when applied to a strongly nonlinear, periodic problem. We fix the cadence by adding a physically motivated periapsis-coverage floor on top of the statistical search. The resulting campaign recovers all seven orbital parameters from simulated astrometry alone, including the injected precession rate: 0.2315 ± 0.0011 deg yr^{-1}, against a true value of 0.2307 deg yr^{-1}. This precision figure assumes a perfect astrometric reference-frame tie, a real instrumental systematic. Section 4.2 shows this systematic must be fit jointly with the orbit, not assumed away, at a final cost of $\sim$2.5$\times$ in precision. This is a synthetic feasibility forecast, not a claim that S301's precession has already been observed. The real next periapsis passage, which such a measurement would require, does not occur until 2031.8. We conclude that a two-passage, astrometry-only GRAVITY+ campaign would be both feasible and sufficient for a spin-independent test of general relativity near Sgr A*—a test of the mass-only (Schwarzschild) precession that holds regardless of the black hole's spin, distinct from the harder Lense–Thirring (spin) measurement S301's discovery motivated. This conclusion holds once the 2031.8 passage and the following passage have actually been observed.

Keywords: Galactic Center — black hole physics — astrometry — celestial mechanics — general relativity — experimental design

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Table 1. Adopted S301 parameters (GRAVITY Collaboration et al. 2026), Solution 1.

Parameter	Value	Uncertainty
Period P	8.680 yr	0.110 yr
Eccentricity e	0.9832	0.0010
Inclination i	124.09°	1.10°
Node Ω	73.8°	3.5°
Argument of periapsis ω	293.4°	2.2°
Periapsis epoch t_0	2023.126 yr	0.010 yr
Semi-major axis a	83.0 mas (687 AU)	···
Sgr A* mass $M_\bullet$	$4.297 \times 10^6\ M_\odot$	···
Distance R_0	8277 pc	···
K magnitude m_K	19.3	0.3

1. INTRODUCTION

The Galactic Center's S-stars have delivered some of the strongest strong-field tests of general relativity to date. These tests include gravitational redshift and Schwarzschild precession in the orbit of S2 (GRAVITY Collaboration et al. 2018, 2020). S301 (GRAVITY Collaboration et al. 2026) extends this program. S301 has a shorter period than S2 (8.68 yr vs. S2's ∼16 yr), a higher eccentricity than S2, and a closer periapsis than S2 (136 R_S). S301 is explicitly proposed as a spin (Lense–Thirring) probe for Sgr A*. A companion paper (T. Piran et al. 2026) develops the methodology for that harder, spin measurement.

Before attempting a spin measurement, a more basic question needs its own answer. Can a real, currently executable observing campaign detect the ordinary, non-spin apsidal precession that any Schwarzschild black hole already produces? We answer this question by building the complete simulation pipeline that a real observing proposal would need. This pipeline has four steps: validate the physics model, design the observing campaign, cross-check that the physics survives realistic noise, and fit the orbit. Two of our own initial design choices failed informative, real tests along the way. We report what worked, what failed, and why (Sections 3–4).

2. THE S301 ORBITAL SOLUTION

All ground-truth parameters are taken from GRAVITY Collaboration et al. (2026), Extended Data Table 2, Solution 1 (Table 1); a second, nearly degenerate solution is also published but not used here. At periapsis S301 passes 136 R_S from Sgr A* at >8% c (∼25,000 km s^{-1})—closer and faster than any other known star, including S2. Its effective temperature is not published; we adopt 7300 K from standard F1.5V tables, affecting only the absolute-magnitude estimate in Section 3.1, not the orbital dynamics used elsewhere.

3. SIMULATION PIPELINE

3.1. *Relativistic Light Curve*

We model S301's K-band light curve using a full Keplerian solution. The model combines the exact special-relativistic Doppler factor (not a small-v approximation) with the Schwarzschild redshift $\sqrt{1 - R_S/r}$. We apply this combined redshift to a blackbody spectrum via the relativistic-beaming relation I_ν/ν^3, evaluated over GRAVITY's real bandpass on the VLTI (Very Large Telescope Interferometer, 1.98–2.40 μm; GRAVITY Collaboration et al. 2017). Four effects are deliberately left out, each for a specific reason. Gravitational lensing (light bending by Sgr A*'s spacetime curvature, which can magnify or split a background source into multiple images) would require full strong-field ray tracing; G. F. Rubilar & A. Eckart (2001) show it is negligible for S2's much wider periapsis (∼1500 R_S), but we have not independently verified that conclusion at S301's far closer 136 R_S, so we flag this as a genuine, unresolved limitation rather than assume it carries over. Occultation does not apply here: nothing physically blocks S301's light along our line of sight, since Sgr A* is a compact source rather than an extended eclipsing body. Kerr (spin-dependent) terms are excluded for the same reason spin is excluded everywhere else in this paper: including them would require

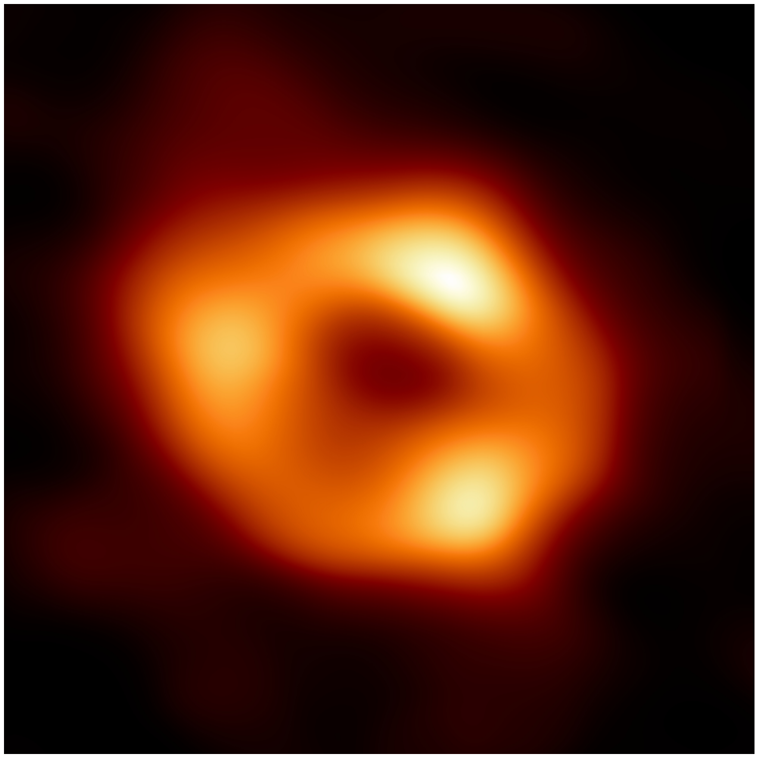

Figure 1. The real Sgr A* itself, at the opposite end of the size scale from everything else in this paper: the Event Horizon Telescope's horizon-scale image of the black hole's shadow, ring diameter 51.8 ± 2.3 μas (Event Horizon Telescope Collaboration 2022)—roughly 1600× smaller than S301's 83 mas orbit (Table 1) and far below any astrometric resolution used in this campaign. Shown for orientation, not analysis. Image credit: EHT Collaboration, released by the European Southern Observatory (ESO) as eso2208-eht-mwa (2022-05-12), licensed CC BY 4.0.

assuming a value for Sgr A*'s unmeasured spin. Rømer (light-travel-time) delay is real and non-negligible—Section 4.5 quantifies its effect directly—but that quantification concerns the astrometric orbit fit's timing, not this light curve, which only needs to validate against the published solution's summary numbers below, not a real fit. None of these four effects has a counterpart in T. Piran et al. (2026)'s analysis to compare against: their method is purely astrometric (multi-star nodal-precession fitting), with no photometric or light-time/timing model at all, so lensing, occultation, Kerr light-bending, and Rømer delay simply do not arise in their treatment—not a choice they made explicitly, but a consequence of asking a different kind of question. The model reproduces those numbers: derived $a = 686.7$ AU vs. 687.0; periapsis 136.0 R_S vs. 136; periapsis speed 8.54% c vs. ">8%". This agreement validates the light-curve engine before we use it to predict anything new.

The predicted periapsis signature is not a symmetric absorption dip. No mechanism in this model produces a dip. Instead, the model predicts an *asymmetric* relativistic-beaming spike. As S301 approaches at high velocity, its brightness rises by $\sim$0.075 mag. As S301 recedes, its brightness falls to $\sim$0.04 mag fainter than baseline, from de-boosting combined with gravitational redshift; redshift dims the star regardless of whether it is approaching or receding. The resulting peak-to-trough amplitude is 0.116 mag (Figure 3). Near periapsis, the total relativistic redshift exceeds the naive classical Doppler prediction by 0.26–0.46 percentage points. This is a genuine, non-negligible correction, not a relabeled classical effect.

This dimming is worth being precise about, because the intuitive picture—Sgr A*'s intense gravity "pulling away" some of S301's light near periapsis—is not quite what this model predicts, though it is close. The net fading after periapsis (Figure 3) does come from gravity acting on the light itself: gravitational redshift genuinely removes energy from each photon climbing out of Sgr A*'s potential well, on top of the special-relativistic de-boosting from S301's own high recession speed. What this model does *not* include is gravitational lensing bending S301's light away from our line of sight, or the black hole capturing photons outright. Those would be real, separate effects; as noted above, we have not verified they stay negligible at S301's periapsis the way they do for S2's wider one.

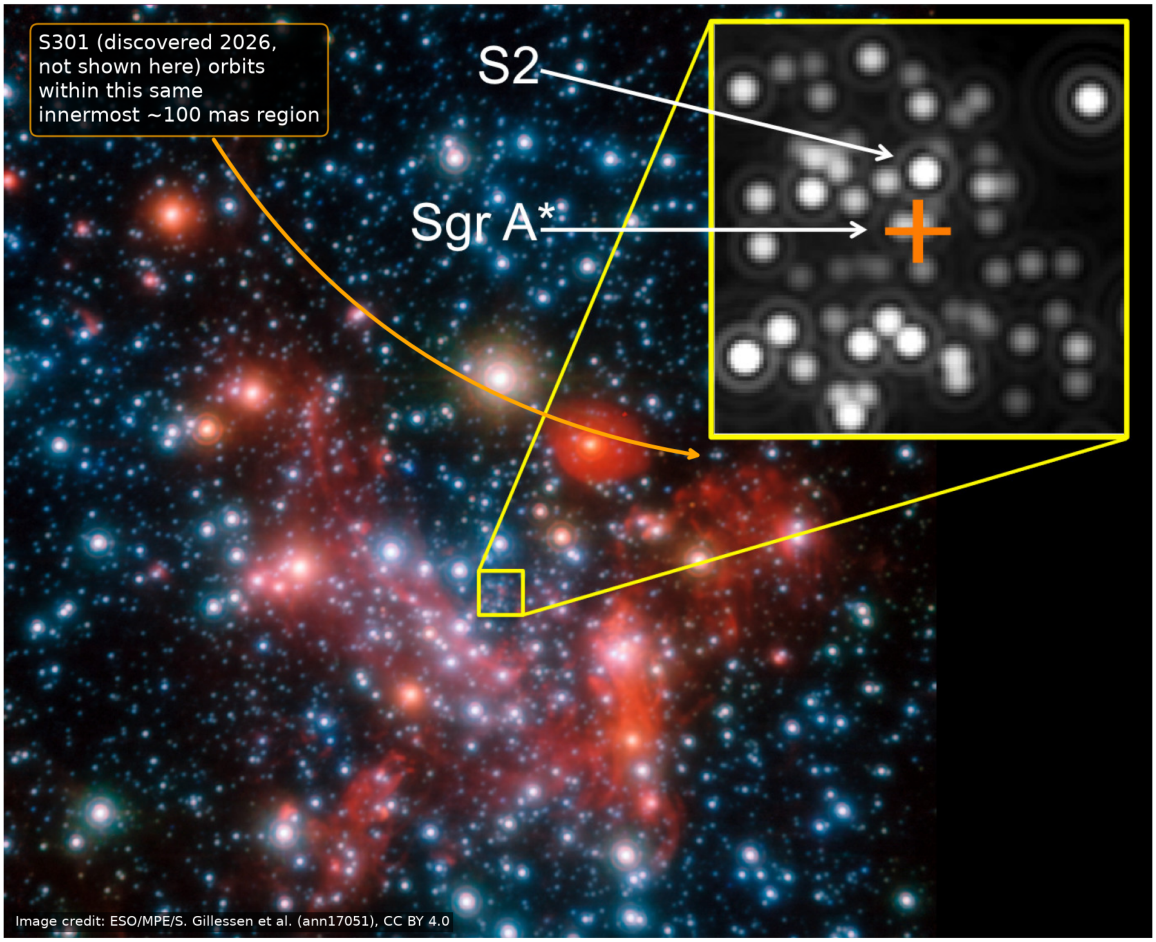

Figure 2. Real near-infrared observational context (not a simulation): the crowded innermost region of the Galactic Center, with the genuine S2/Sgr A* zoom inset from the original release. S301 was discovered in 2026, after this 2017 image, and is not itself resolved in it; the annotation is schematic, pointing out that S301's orbit (semi-major axis 83 mas, comparable to S2's) occupies this same innermost few-tens-of-mas region, illustrating the source crowding relevant to Section 4.1's discussion of instrumental confusion. Image credit: ESO/MPE/S. Gillessen et al., released as ann17051 (2017-08-09), licensed CC BY 4.0.

3.2. *Observing Campaign: Instrument and Cadence*

The campaign uses GRAVITY+, the real VLTI interferometer that made the S301 discovery. GRAVITY+ combines the light from the four 8.2 m Unit Telescopes at Paranal into a single interferometric beam, using adaptive optics on each telescope to sharpen the incoming starlight first. Combining four telescopes this way gives it an effective resolving power far beyond any single telescope's, which is how it reaches milliarcsecond-to-sub-milliarcsecond precision in the crowded, faint Galactic Center field. We adopt its demonstrated 100 μas astrometric precision (GRAVITY+ Collaboration 2023) and its simultaneous K-band photometry.

Radial velocity was considered and rejected before any data was simulated. SINFONI, an integral-field spectrograph formerly at the VLT (Very Large Telescope), delivered a real, achieved S2 radial-velocity precision of 12.3 km s^{-1} at $m_K = 14.0$ (GRAVITY Collaboration et al. 2018); we scale that benchmark to S301. ERIS, the VLT's newer instrument, replaces SINFONI with SPIFFIER, its own near-infrared integral-field spectrograph, at resolving power $R = 5000$ (R. I. Davies & et al. 2023). At S301's magnitude ($\Delta m = 5.3$ fainter than S2, $\sim$130$\times$ less flux) and under the background-limited noise regime appropriate for faint Galactic Center spectroscopy, the SINFONI benchmark scales to a velocity precision of $\sim$1621 km s^{-1}. Two separate ratios matter here, and they are easy to conflate. First, precision against signal: S301's RV signal is $\pm$15,000 km s^{-1}, so precision-to-signal is 1621/15,000 $\approx$ 0.11, i.e. each single-epoch measurement carries roughly 9–13$\times$ more noise than signal. Second, and separately, precision against

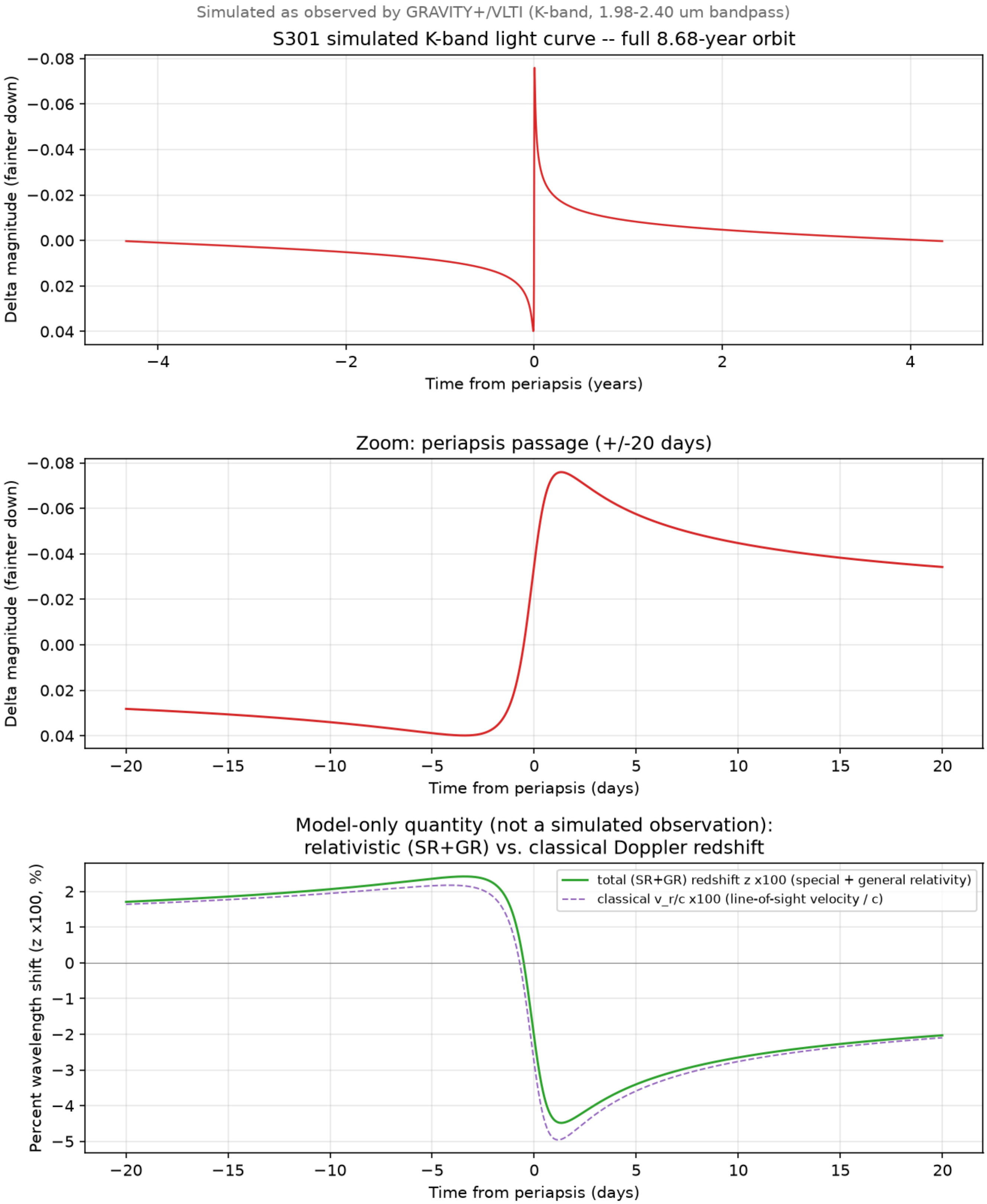

Figure 3. Simulated relativistic K-band light curve of S301 over one orbit (top), a zoom on the periapsis passage (middle), and the relativistic vs. classical spectral shift near periapsis (bottom). The middle and bottom panels show a generic passage, in time measured from periapsis itself; the shape is the same for either real passage (2031.8 or 2040.5), since precession changes the orbit's orientation on the sky, not its brightness or redshift as a function of time from periapsis. The bottom panel is a theoretical, model-internal quantity (SR = special-relativistic Doppler shift, GR = Schwarzschild gravitational redshift), not a simulated observation: no instrument in this campaign measures a spectral shift directly (Section 3.2); it is shown to explain what drives the delta-magnitude curves above it.

SPIFFIER's own 60 km s^{-1} resolution element: $60/1621 \approx 0.037$, the fraction of one resolution element that this measurement could hope to resolve. This second ratio is the one a Time Allocation Committee actually asks about, because it says whether the line centroid can be pinned down at all within one spectral resolution element; a value this far below 1 means no. No committee grants ERIS time for a measurement known in advance to fail this badly, so this campaign never requested ERIS radial-velocity time. We confirmed this directly, not by assumption: refitting identical astrometric data with a simulated RV channel added back leaves every parameter's bootstrap uncertainty unchanged, including the precession rate (0.0011 deg yr^{-1} either way). RV adds no useful constraining power to the orbit fit at S301's magnitude.

Getting the cadence right took three attempts. S301 spends nearly all of its 8.68 yr period near apoapsis. Almost all of its angular motion happens within days of periapsis. Cadence, the schedule of when the instrument observes, therefore matters enormously: where the observation epochs land in time strongly determines what the orbit fit can recover. Design A, described below, is a straightforward first guess at a good cadence. Designs B and C are two attempts to beat Design A using rigorous statistical optimization. Both attempts fail, for related reasons explained below. Design D is the third attempt, and the one used in this study. Design D succeeds by adding a non-statistical constraint on top of the same optimization method that failed for Designs B and C.

Designs B and C both rely on Fisher information, a concept from experimental design theory. Fisher information answers a concrete question about a candidate observation: an epoch when GRAVITY+ measures S301's position on the sky. The question is how much that epoch is predicted to shrink the eventual uncertainty on the fitted orbital parameters, such as the period, the eccentricity, and the precession rate. A *D-optimal* design chooses epochs to jointly shrink the uncertainty on all seven fitted parameters at once, rather than optimizing each parameter one at a time. Concretely, a D-optimal design picks the set of epochs that together leave the smallest possible combined uncertainty region for all seven parameters. A smaller combined uncertainty region means a more informative cadence.

Design A (heuristic): the obvious first guess. Periapsis is where S301's orbit changes fastest, so a naive cadence should observe there often. ~90% of epochs fall within ±20 days of each periapsis; the remaining epochs are sparse. Design A is simple and reasonably effective (Table 2). Design A assumes, without proof, that all seven fitted parameters are equally well served by periapsis proximity.

Design B (naive D-optimal): the statistically "optimal" approach on paper. Instead of guessing where to observe, Design B lets Fisher information pick the epochs, with no human assumption about periapsis. It computes Fisher information at one fixed, assumed-true set of parameter values, then greedily chooses epochs to maximize the resulting joint precision. Design B **failed catastrophically**. It placed nearly the whole observing budget on two clusters at the campaign's temporal extremes, 5.4 and 18.5 yr from the reference epoch and more than two orbital periods away. It selected *zero* epochs near either periapsis. Its Cramér–Rao prediction promised uncertainties 1.8–28,000× tighter than Design A's. A real bootstrap fit instead came out 4–600× *worse* than Design A on every parameter.

Here is why Design B failed. Sensitivity to the orbital period genuinely grows with the time baseline between observations; the same effect makes pulsar timing so powerful over long baselines. But that growth is a *local* property of the linearized model: it holds only near the true parameter values where Fisher information was evaluated. Extrapolated across more than two orbital periods, far from where it was evaluated, a tiny period error compounds into a large phase error. This is a cycle slip: the predicted orbital phase drifts by a large, wrong amount even though the period error that caused it was small. A clock that runs a few seconds fast per day shows the same effect. After one day, the clock is barely wrong. After a year, it is off by half an hour. Linearized Fisher information, evaluated only near the assumed true parameter values, cannot see this nonlinear blow-up far from where it was evaluated.

Design C (pseudo-Bayesian robust): a more robust version of Design B, hedging against not knowing the true orbital parameters in advance. Rather than computing Fisher information at one fixed guess, Design C averages the calculation over 300 draws from a realistic parameter prior (K. Chaloner & I. Verdinelli 1995). This averaging makes Design C less sensitive to being wrong about the true orbit, and it de-emphasized the fragile temporal-extreme clusters that doomed Design B. But Design C *still* selected zero periapsis epochs. The reason is structural, not statistical. Fisher information, even averaged over a prior, measures only noise-driven statistical uncertainty. It cannot represent geometric non-identifiability: the failure that comes from sampling too few distinct orbital phases, no matter how precisely each one is measured. This is exactly why classical orbit determination has always required observations spread across an orbit's curvature, not just wherever a statistical criterion points.

Design D (constrained): give up on asking pure statistics to find periapsis coverage on its own, and require that coverage by hand instead. A hard, non-statistical floor reserves ~30% of the observing budget within ±20 days of each periapsis. The remaining ∼70% is still allocated by the same Fisher-optimal search used in Designs B and C. Design D is the design used in this study, and it beats Design A on every parameter (Table 2). Design D wins because it combines domain knowledge with statistical optimization instead of picking one or the other. Periapsis coverage matters, full stop, and the hard floor guarantees it; neither purely statistical design, B nor C, found that coverage on its own. Fisher-optimal allocation of the remaining budget then extracts extra precision from every epoch that the heuristic Design A does not know how to place.

This single-star cadence problem is a different optimization from T. Piran et al. (2026)'s multi-star differential design, which must additionally synchronize observations across several stars to statistically subtract out Newtonian confusion when isolating the spin (Lense–Thirring) signal. Here, with one star and one observable, apsidal precession, to track, the constraint that matters is periapsis coverage in time, not cross-star simultaneity.

All four designs are filtered to Sgr A*'s real Paranal visibility season (∼March–September; GRAVITY Collaboration et al. 2026). This filtering exposes a real scheduling gap, and it is more severe than a simple anchoring adjustment. We checked Sgr A*'s actual solar elongation across the calendar year: it drops below 45°, effectively unobservable by any facility, from 2031 November 6 to 2032 February 1, bottoming out at 5.6° on 2031 December 19. The first periapsis, 2031 October 22, falls squarely inside this conjunction. Its dense observing window is anchored to the nearest visible date, ∼2031 September 30, roughly 22 days before periapsis, and the earliest the field reopens afterward is ∼2032 February 1, roughly 102 days after periapsis. This campaign does not merely place its window suboptimally around passage 1; it cannot observe periapsis itself, or the first three-plus months of the recovery that follows it, at all. This is a hard limit of viewing Sgr A* near solar conjunction, not a cadence choice, and it is not specific to Paranal or to astrometry. Sgr A*'s elongation stays below 85°, JWST's own minimum sunshield-driven pointing limit, from 2031 September 26 through 2032 March 12, so a space telescope at Earth-Sun L2 gains nothing here; HARMONI's site (Cerro Armazones) is 20 km from Paranal and shares the identical sky; and the constraint is geometric (pointing too close to the Sun), not tied to the observing technique, so a spectrograph is no more able to see through it than an imager. Only a spacecraft at a genuinely different heliocentric vantage point could evade this specific conjunction window, and no real or concretely planned mission offers that for Sgr A*. The second passage, 2040 June 26, falls comfortably in-season and is not subject to any of this.

3.3. *Photometric Cross-Check*

GRAVITY+'s astrometric observations produce photometry as a free byproduct. Every astrometric epoch also yields a K-band brightness measurement, at no extra cost in telescope time. We generate this photometry. We do not fit it.

Astrometry alone already fully determines S301's orbital state, its position and velocity, at every epoch, through the same seven parameters this paper fits: the six Keplerian elements, plus the precession rate $\dot{\omega}$ (Section 3.4). Photometry is then just a fixed, computable function of that same state, via Section 3.1's relativistic light-curve model, so it introduces no new free parameter. In the idealized limit of perfect, noise-free photometry, this would not mean photometry carries *zero* information: two independent measurements of the same underlying state generally do combine to shrink the joint uncertainty, exactly as Section 5's HARMONI-RV discussion shows for a genuinely well-measured radial velocity. Whether that theoretical extra information matters in practice is a separate, testable question, and we tested it directly rather than assuming an answer either way. We refit the same 131-epoch dataset with a simulated photometric channel added to the astrometric residuals, jointly constraining all seven parameters, and bootstrapped the result exactly as for the astrometry-only fit. Every parameter's precision is unchanged to within ordinary bootstrap noise, including $\dot{\omega}$ (0.0011 to 0.0012 deg yr^{-1} either way). At S301's real photometric uncertainty (0.30 mag against a 0.116 mag signal), the photometric channel's Fisher-information contribution is real but negligible next to astrometry's, not exactly zero as a cruder argument would claim, but small enough that fitting it jointly buys nothing measurable.

That is not the same question as whether photometry is worth generating at all. It is, for a different reason: an independent validation check.

The photometric epoch count is not an independent design choice. It is fixed at 131, the same astrometric cadence Section 3 designs (60 epochs for one passage, doubled to 131 for two), because every photometric point rides along with an astrometric one for free. An earlier, space-based version of this project considered Roman (the Nancy Grace Roman Space Telescope), which had continuous, season-independent access and a much larger single-passage epoch count under discussion; that design was abandoned once Roman's 2.4 m aperture proved unable to resolve S301's milliarcsecond-scale orbit at all (Section 5). The current ground-based GRAVITY+ design's cadence was rebuilt independently around a realistic single-passage baseline, arriving at 60 epochs, then 131 for two passages (Section 3).

A reader might still ask why the campaign does not schedule far more photometric epochs, dense enough to average the noise down further. Two honest reasons apply. First, GRAVITY+ is a shared, oversubscribed facility on a real telescope, not a source of free continuous monitoring. Nightly-for-40-days coverage of one target would compete with every other approved Paranal program. The dense window this paper actually uses, roughly one measurement every few days near each periapsis rather than every night, already reflects a realistic cadence, not a maximal one. Second: the joint-fit test above shows the photometric channel is already far too noise-dominated, at S301's real magnitude, to move $\dot{\omega}$'s precision measurably with the epoch count this campaign already has. More photometric epochs, drawn from the same 0.30 mag-per-point noise floor, would sharpen the validation check below, not this paper's science result, for the same reason: the astrometric channel's information content already dwarfs photometry's by a wide margin, and adding more low-information epochs closes that gap only very slowly. This paper does not derive a first-principles optimal photometric cadence, the way Section 3 derives the astrometric cadence from Fisher information. We say so plainly, rather than imply a precision this paper does not have.

S301's published photometric uncertainty (0.30 mag) exceeds the entire 0.116 mag signal from Section 3.1 at any single epoch. Figure 4's bottom panel shows the consequence directly: the simulated data points scatter around the injected truth curve at a level consistent with that 0.30 mag uncertainty. No epoch's scatter is anomalously large or small. The gray truth curve drawn in that panel is the same curve the simulation injects, not a curve fit to the data afterward. On its own, this confirms the noise model, the injected light-curve physics, and the observing cadence are mutually consistent, end to end. It is not, by itself, a claim that the periapsis brightening feature is visually detectable in this photometry; it is not detectable by eye in this dataset, at S301's magnitude.

A stronger, independent check is possible, and we ran it: take the orbital solution from the astrometry-only fit (Table 3, no photometry involved), use it to *predict* the brightness curve via Section 3.1's physics, and compare that prediction against the simulated photometric data, which the astrometric fit never saw. This is the genuine test of whether the beaming-and-redshift model this paper is built on, the effect that motivated looking at S301's light curve in the first place, actually conforms to independent data, rather than a truth-curve-against-itself sanity check. The result: $\chi^2 = 142$ over 131 epochs (reduced $\chi^2 = 1.08$), statistically consistent with a correct model. An orbit derived purely from position measurements correctly predicts an independent brightness measurement it was never fit to. That is a real, if modest, validation of the light-curve physics, obtained without paying any cost in $\dot{\omega}$ precision, exactly because photometry was used to check the model rather than to fit it.

3.4. *Orbit Fit*

We fit seven parameters to the astrometric data: the six Keplerian elements, plus the apsidal precession rate $\dot{\omega}$. We apply $\dot{\omega}$ as $\omega_{\rm eff}(t) = \omega + \dot{\omega}\,(t - t_{\rm ref})$. The fit uses Levenberg–Marquardt least squares. Its initial guess starts from S301's published discovery-paper orbital solution (GRAVITY Collaboration et al. 2026), not from a blind search. S301 already has a preliminary orbital solution in hand. A real follow-up campaign would not need a periodogram or any other from-scratch period search; it would start from that published solution and refine it with new data. This fit's initial guess simulates exactly that scenario. We take the published solution and perturb it by a fixed offset, standing in for realistic uncertainty in a preliminary orbit rather than a from-scratch determination: 3% in period, 0.03 in eccentricity, 10° in inclination, 15° in longitude of the ascending node, 12° in argument of periapsis, 0.05 yr in time of periapsis passage, and 0.1 deg yr^{-1} in the precession rate. These perturbations are not trivially small. At each fit iteration, we derive the semi-major axis from $(P, GM_\bullet)$ via Kepler's third law. Parameter uncertainties come from 1000 bootstrap resamples. We compare the recovered $\dot{\omega}$ to the analytic Schwarzschild prediction,

$$\dot{\omega} = \frac{6\pi\, GM_\bullet}{c^2\, a\,(1-e^2)\, P}, \tag{1}$$

the Mercury-perihelion formula applied to S301's orbit. We inject no spin (Lense–Thirring) term anywhere in the simulation: the inclination i and node Ω remain genuinely static throughout. We do this because Sgr A*'s spin is unmeasured; assuming a spin value would substitute a guess for a real measurement.

4. RESULTS

The final campaign has 131 epochs (2028.50–2041.69). Of these, 50 (38.2%) fall within ±20 days of a periapsis, comfortably above Design D's 30% hard floor (Section 3.2): the Fisher-optimal search chose to place some of its own

Table 2. Bootstrap 1σ uncertainties, heuristic vs. constrained cadence ($N = 131$, RV included to isolate cadence).

Parameter	Design A	Design D	Published
P (yr)	0.0001	0.0002	0.1100
e	0.0001	0.0000	0.0010
i (deg)	0.0476	0.0351	1.1000
Ω (deg)	0.1214	0.0814	3.5000
ω (deg)	0.0932	0.0459	2.2000
t_0 (yr)	0.0003	0.0003	0.0100
$\dot{\omega}$ (deg yr^{-1})	0.0043	0.0011	⋯

flexible budget near periapsis too, on top of the guaranteed floor. This 131-epoch, two-passage campaign is more than double an earlier 60-epoch single-passage design.

The remaining 81 epochs (61.8%) are not spread evenly across the 13.2 yr baseline. They land in a small number of tight clusters, separated by multi-year gaps with no epochs at all (visible directly in Figure 4's epoch-colored track), including one at 2033 and another spanning 2037–2038, rather than immediately after passage 1 once the conjunction described above releases the field in 2032. This looks wasteful, and we tested it directly, in two different ways, rather than defending it by argument alone.

The first, more fundamental question is whether the conjunction gap itself costs anything. We built a hypothetical design, identical to Design D except that passage 1's dense cluster is a complete, symmetric ±20-day window centered on true periapsis, as if the conjunction did not truncate it to the 22 days of pre-periapsis approach this campaign is actually limited to, and bootstrap-fit both. Completing the passage this way tightens P and t_0 substantially (39% and 51%) and i, Ω, ω modestly (~3–4%), while leaving $\dot{\omega}$ statistically unchanged (0.00115 vs. 0.00114 deg yr^{-1}). So the conjunction gap is a real loss for several parameters, and not a hidden blessing for any of them; it costs nothing measurable on $\dot{\omega}$ specifically only because that parameter's precision is dominated by the ~8.7 yr baseline between the two periapsis-anchored clusters, a baseline this campaign already uses close to fully, not because incomplete periapsis coverage is somehow advantageous.

The second, separate question is what to do with the budget freed up by that unavoidable gap. Here we tested moving the 2033 and 2037–2038 flexible epochs into the nearest window the conjunction actually permits (2032.09–2032.74, starting 103 days after periapsis, well into the slow coast rather than the fast part of the passage) instead. That specific reallocation makes $\dot{\omega}$'s precision *worse* by 46% (0.0011 to 0.0017 deg yr^{-1}), and every placement we tried between 2032 and 2037 followed the same pattern. The reason is not that avoiding periapsis helps; it is that epochs placed immediately after passage 1, in the slow coast the conjunction leaves accessible, sample a state dynamically close to what the existing dense cluster already measures well, and are largely redundant with it. Epochs at a genuinely different orbital phase instead help resolve the remaining parameter degeneracies (period against epoch of periapsis, node against inclination) that would otherwise blur the comparison between the two periapsis anchors. A D-optimal search picks epochs to maximize joint information for a periodic signal, and here that means a few widely-separated, tightly-grouped clusters at specific phases, not proximity to periapsis for its own sake once a periapsis is already well anchored elsewhere. The result serves this paper's actual goal, recovering seven orbital parameters from astrometry, but it is not designed to trace out the light curve continuously in time; Section 3.3 addresses that trade-off directly.

Design D improves i, Ω, ω, and $\dot{\omega}$ by 26–74%, relative to Design A, at negligible cost to P/t_0 (both remain >100× tighter than published uncertainty regardless, for either design).

Every parameter recovers within $\sim 1.5\sigma$ of truth. The fitted sky-plane track (Figure 4, top panel) visibly shows the periapsis direction rotated between passages—the geometric signature of precession, closing the loop opened in Section 3.1. Because this campaign assumes best-case GRAVITY+ precision at every epoch across two full passages 8.68 yr apart (a baseline the real 2026 dataset does not yet have), several uncertainties here are tighter than the real

Table 3. Final result: Design D cadence, astrometry only.

Parameter	Truth	Fitted	σ	Published σ
P (yr)	8.6800	8.6800	0.0002	0.1100
e	0.9832	0.9832	0.0000	0.0010
i (deg)	124.0900	124.1123	0.0351	1.1000
Ω (deg)	73.8000	73.7284	0.0814	3.5000
ω (deg)	293.4000	293.3615	0.0459	2.2000
t_0 (yr)	2023.1260	2023.1261	0.0003	0.0100
$\dot{\omega}$ (deg yr^{-1})	0.2307	0.2315	0.0011	$\cdots$

paper's published values (e.g., 0.08° vs. 3.5° on Ω)—an idealized-study artifact, not a claim of outperforming the real discovery.

A further, non-circular consistency check strengthens this result. $\dot{\omega}$ above is a free 7th parameter, fit purely from geometry with no reference to Equation 1. For each bootstrap resample we instead compute what that formula predicts using *that same resample's own fitted* (P, e)—still blind to any published or injected value, since a real discovery would have no such value to check against either. This analytic, fit-derived prediction is 0.2312 ± 0.0005 deg yr^{-1}, and its per-resample difference from the independently-fit $\dot{\omega}$ is 0.0002 ± 0.0014 deg yr^{-1} (0.17σ from zero, using the per-resample difference rather than a naive combination of the two marginal uncertainties, since P, e, and $\dot{\omega}$ are drawn from the same correlated resamples). The geometry and the theory agree without either one referencing the other's answer.

4.1. *Systematic Error Budget: Extended-Mass Confusion*

The uncertainties in Table 3 are random (photon-noise-driven) only, from bootstrap resampling of simulated measurement noise. A comprehensive error budget also requires astrophysical and instrumental systematics. We quantify the leading astrophysical one directly: any real, undetected extended ("Newtonian confusion") mass near Sgr A* retrogradely shifts the periapsis and biases a measured apsidal precession low, an effect first modeled quantitatively by G. F. Rubilar & A. Eckart (2001) for S2-like orbits.

We reproduce their post-Newtonian treatment exactly, not a re-derived approximation: a spherically symmetric matter distribution combining Sgr A*'s point mass with a Plummer-like component, $\rho_\alpha(r) = [1 + (r/r_c)^2]^{-\alpha/2}$ ($\alpha = 5$, the classical Plummer profile), integrated numerically via their Eq. A.7 post-Newtonian equation of motion, $\dot{\mathbf{v}} = -GM(r)r^{-3}[(1 + 4\phi/c^2 + v^2/c^2)\mathbf{r} - 4(c^{-2})\mathbf{v}(\mathbf{v} \cdot \mathbf{r})]$, with $M(r)$ and $\phi(r)$ the exact enclosed-mass and potential integrals (their Eqs. 12–13) for the assumed profile. We integrate S301's real orbit through two periapsis passages (`extended_mass_error.py`) and measure the eccentricity-vector rotation between them—a purely geometric, astrometric quantity requiring no radial-velocity data. The point-mass-only case validates the integrator against Equation 1 to 1.2%, consistent with the $(v/c)^2 \approx 0.7\%$ second-post-Newtonian scale expected at S301's $>8\%c$ periapsis speed; because the extended-mass bias below is a *difference* between two runs of the identical integrator, this common-mode residual mostly cancels.

For the extended-mass magnitude and core radius, we use GRAVITY Collaboration et al. (2022)'s own multi-star constraint self-consistently—their fiducial Plummer scale length ($0.3'' \approx 12.0$ mpc at S301's distance) together with their own mass bound, rather than pairing a current bound with G. F. Rubilar & A. Eckart's decades-old fiducial r_c. That analysis limits any extended mass within S2's apocenter to $\lesssim 3000\ M_\odot$ (1σ) or $\lesssim 7500\ M_\odot$ (their conservative 3σ limit). At the 1σ bound, the resulting bias on S301's apsidal precession is -0.00013 deg yr^{-1} (0.06% of the Schwarzschild signal, 0.09σ of the fitted bootstrap precision in Table 3); at the conservative 3σ bound, -0.00032 deg yr^{-1} (0.14%, 0.23σ). Even at the least favorable real bound, extended-mass confusion is subdominant to random measurement error for this campaign—because S301's periapsis (0.056 mpc) sits so far inside any plausible $r_c \gtrsim$ few mpc that only a small fraction of the already-small bounded extended mass is enclosed there, unlike S2's more modest relativistic signal, which the same confusion mass more readily competes with.

This estimate is necessarily narrower than T. Piran et al. (2026)'s treatment of the same underlying hazard: they model a flattened disk's *nodal* (Ω) precession competing with the Lense-Thirring spin signal, using multi-star differential

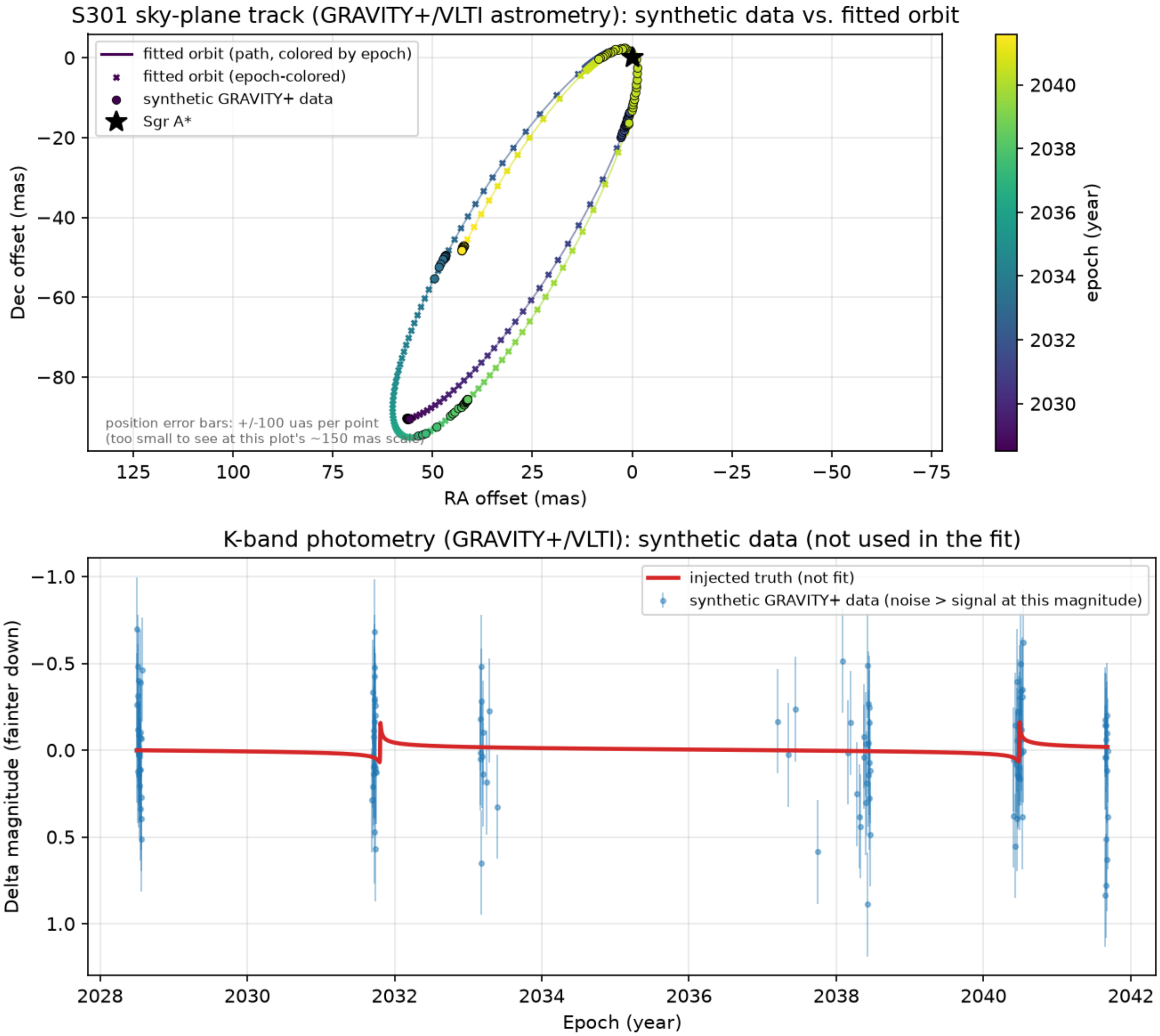

Figure 4. Top: fitted sky-plane track (GRAVITY+/VLTI astrometry) vs. synthetic data, epoch-colored, showing the precessing orbit's two passages. Bottom: simulated K-band photometry vs. the injected truth curve (not used in the fit). An animated version (`s301_precession.mp4`, included as an ancillary file with this submission and available at https://github.com/jcatanza/s301-schwarzschild-precession/blob/master/output/s301_precession.mp4) extrapolates the same real 1PN precession rate over 20 orbits (∼174 yr) to make the rosette pattern visible; only the two orbits actually analyzed in this paper are highlighted there, and the extrapolation is labeled explicitly on-screen throughout.

subtraction; we model a spherical distribution's contribution to the *apsidal* (ω) precession studied here, for S301 alone. The two are not directly comparable numbers, but they address the same qualitative concern for different observables, and our result is consistent with theirs in finding S301 comparatively well-protected against this systematic (their Section 3 reports S301's disk/LT confusion ratio ∼10× better than S2's).

4.2. *Systematic Error Budget: Astrometric Reference-Frame Tie*

The leading instrumental systematic is the astrometric reference-frame tie: S301's position is measured in an infrared (adaptive-optics/interferometric) frame that must be registered to Sgr A*'s true dynamical center via an independent radio (VLBI) astrometric solution, which itself carries a residual zero-point offset and a slow linear drift. This is not hypothetical—it is the same systematic GRAVITY Collaboration et al. (2020) identify as limiting their own S2 measurement: "these reference frame parameters are now the limiting factor in the precision of the detection of the SP of S2," adopting frame-tie priors from P. M. Plewa et al. (2015) of $x_0 = -0.2 \pm 0.2$ mas, $y_0 = 0.1 \pm 0.2$ mas, $\dot{x}_0 = 0.05 \pm 0.1$ mas yr^{-1}, $\dot{y}_0 = 0.06 \pm 0.1$ mas yr^{-1}. GRAVITY Collaboration et al. (2022) state explicitly that this frame drift and apsidal precession "are degenerate"—i.e., an unmodeled frame tie does not just add noise, it can be reinterpreted by the fit as spurious precession.

Table 3's pipeline has no frame-tie parameters at all—a realistic first-pass analysis that implicitly assumes a perfect tie. We test this directly (`instrumental_error.py`): inject the real P. M. Plewa et al.-magnitude drift into the existing synthetic dataset and refit with the unmodified 7-parameter model. The result is severe: at the central Plewa et al. values, $\dot{\omega}$ is biased by -0.0377 deg yr^{-1}—27σ relative to the random-error precision in Table 3, 16% of the entire signal; at a conservative 1σ-worse realization, -0.107 deg yr^{-1} (77σ, 46% of the signal). Unmitigated, this single instrumental term dwarfs both the random error and the extended-mass astrophysical systematic (Section 4.1) by more than an order of magnitude.

The fix is the one GRAVITY Collaboration et al. (2020) themselves use: fit the frame-tie parameters jointly with the orbit, constrained by their real Gaussian priors, rather than assuming them away. Extending the model to 11 parameters ($x_0, y_0, \dot{x}_0, \dot{y}_0$ added, Gaussian-penalized against the P. M. Plewa et al. priors) and refitting the same drift-injected data recovers $\dot{\omega}$ to within 0.3% of truth in both cases—the frame parameters themselves come back consistent with their injected values to within the prior width. Propagating this properly (bootstrap over the 11-parameter fit) gives a final $\dot{\omega}$ precision of 0.0028 deg yr^{-1}, $\sim 2.5\times$ wider than Table 3's frame-perfect value (0.0011 deg yr^{-1}) but still two orders of magnitude better than the naive fit's unmitigated bias. As with the cadence design (Section 3), a naive approach fails badly and invisibly, while explicitly modeling the known physical/instrumental degeneracy—using real, published priors rather than assuming a perfect instrument—recovers the result.

4.3. *Systematic Error Budget: Source Confusion/Crowding*

The remaining instrumental term is source confusion: any neighboring star within roughly one diffraction-limited beam blends into the same single-mode-fiber output and biases the recovered photocenter by (to leading, instrument-agnostic order) its flux-weighted contribution. We bound this with a Monte Carlo (`confusion_error.py`): random realizations of the local field around S301 ($m_K = 19.3$), drawing neighbor counts and positions from GRAVITY Collaboration et al. (2022)'s own quoted stellar surface density (">100 stars per square arcsecond to $K < 17$," with the number-magnitude slope out to $K = 21$ taken from G. F. Rubilar & A. Eckart's quoted Jaroszyński 1999 ratio, $\nu_{21} \approx 25 \times \nu_{17}$, applied to this updated normalization), weighted by a Gaussian beam-coupling kernel matched to GRAVITY Collaboration et al. (2020)'s quoted 60 mas single-UT full width at half maximum (FWHM, the beam's diameter measured where its sensitivity falls to half its peak value).

Applied naively (a single-aperture imaging photocenter, with no interferometric advantage), this gives a median bias of ~ 9.7 mas per epoch—wildly larger than GRAVITY's real, demonstrated ~ 100 μas precision, confirming that the naive photocenter number is the wrong quantity to use directly. GRAVITY Collaboration et al. (2022) state interferometric astrometry is "much less affected (by a factor of several hundred) by confusion noise" than this naive imaging case; bounding that qualitative range as 200–500×, the actual GRAVITY confusion floor (90th-percentile realization) is 37–92 μas, comparable to or somewhat below the 100 μas photon-noise floor already modeled in Table 3.

Unlike the frame tie (Section 4.2), confusion is not a coherent drift across the campaign's baseline—neighboring S-cluster stars move on their own orbits, so a given blend configuration is not persistent between widely separated epochs. We therefore propagate it as additional per-epoch scatter (quadrature-combined with the existing 100 μas floor) rather than a coherent offset, and re-run the real bootstrap fit with this inflated sigma: even at the conservative end (200× suppression, 90th-percentile realization), $\dot{\omega}$'s bootstrap precision is unchanged to 3 significant figures (0.0012 deg yr^{-1}, a 1.00× ratio), and the best-fit value shifts by $< 10^{-4}$ deg yr^{-1}—consistent with re-optimization noise, not a systematic bias. Confusion/crowding is real but, for this campaign's interferometric technique, subdominant to the already-modeled random error.

4.4. *Beyond the Error Budget: Spin (Lense-Thirring) Contamination*

The three systematics above share a structure: each has a known or boundable magnitude, so each can be modeled, subtracted, or shown subdominant. Sgr A*'s spin is different in kind. For a spin aligned with S301's orbital angular momentum, frame-dragging produces a purely apsidal shift (`orbit.lense_thirring_apsidal_rate`, validated against

GRAVITY Collaboration et al.'s own quoted 0.114 deg/orbit at $\chi = 1$)—i.e., it adds directly and indistinguishably to the same $\dot{\omega}$ this entire pipeline fits for, with no analogous upper bound to inject the way GRAVITY Collaboration et al. (2022)'s extended-mass limit was used in Section 4.1.

Sgr A*'s spin is not robustly measured; real estimates vary by method. Using GRAVITY Collaboration et al. (2026)'s own fiducial benchmark ($\chi = 1$, aligned) and an independent radio/X-ray outflow-method measurement ($\chi = 0.90 \pm 0.06$; R. A. Daly & et al. 2024), the possible Lense-Thirring contamination of $\dot{\omega}$ is 6.0–12.0σ relative to this project's random-error precision (2.9–5.7% of the Schwarzschild signal itself)—larger, in sigma terms, than any of the three bounded systematics after their respective fixes. A measured $\dot{\omega}$ from S301 alone cannot distinguish a pure point-mass GR test from GR plus aligned frame-dragging; this is exactly the degeneracy T. Piran et al. (2026)'s multi-star differential method exists to break, and exactly why this paper's own spin-agnostic framing (Section 5, *Scope*) is a scope choice, not a claim that the reported $\dot{\omega}$ is free of this effect.

4.5. *A Geometric Effect: Roemer (Light-Travel-Time) Delay*

orbit.py's engine explicitly excludes Roemer delay (Section 3.1): the light-travel-time from S301 to the observer varies over the orbit as the star's line-of-sight distance changes, so an observed arrival time does not equal the dynamical emission time the model assumes. Neither Table 3's headline fit nor the synthetic campaign it is built from injects this effect either; that result is self-consistently light-time-naive on both sides (truth and model agree with each other, but neither reflects a real light-travel-time delay a real dataset would have). This section is a separate, standalone test: it injects the real effect into its own synthetic dataset specifically to measure its bias and demonstrate a fix, and that test does not feed back into Table 3's reported number. S301's orbital diameter implies a maximum possible swing of $2a/c \approx 7.9$ days—comparable to the campaign's own $\pm$20-day periapsis-dense window (Section 3); across the real campaign epochs the actual correction ranges -0.06 to $+5.75$ days (root-mean-square, RMS, of 3.48 days).

We tested this directly (`roemer_delay.py`): building a noise-free synthetic dataset with the true light-time correction applied, then refitting with the unmodified (light-time-naive) 7-parameter model. The effect on $\dot{\omega}$ is modest—a bias of -0.0017 deg yr^{-1} (0.75% of the signal, 1.3σ relative to Table 3's random-error precision)—but the effect on the orbit's orientation angles is not: i, Ω, and ω are biased by $+0.49°$, $-1.31°$, and $-0.70°$ respectively, corresponding to 14–16σ relative to their own bootstrap uncertainties. Unlike the other systematics in this budget, Roemer delay barely touches the parameter this paper is built around, while substantially distorting the recovered orbital geometry itself.

The correct fix requires no new free parameters, unlike the frame tie's real nuisance parameters (Section 4.2): light time is a deterministic function of the orbit already being fit, so a self-consistent light-time correction inside the model (solving $t_{\rm emit} = t_{\rm obs} + z(t_{\rm emit})/c$ for each trial parameter vector) is, in principle, exact. We verified this directly at the true parameters—the corrected model reproduces the noise-free data to machine precision (zero residual)—confirming the bias above is a modeling omission, not a data artifact. Robustly *recovering* that optimum via nonlinear least-squares initially failed for an unrelated reason: tracing the failure down to individual observation epochs uncovered a genuine bug in `orbit.py`'s shared Kepler-equation solver, used by every calculation in this project. Undamped Newton-Raphson from the naive starting guess $E_0 = M$—this function's original implementation—diverges catastrophically and permanently for a subset of ordinary (mean anomaly, eccentricity) pairs at S301's $e = 0.9832$ (verified directly: $M = 0.2185$ rad, an unremarkable value, blows up to $E \sim 10^{13}$ and never recovers within the full 200-iteration budget). This had not been visible elsewhere in this project because the true orbital elements' own epochs happen to avoid the divergent region, but a nonlinear optimizer exploring nearby trial parameters—exactly what fitting does—routinely does not. We replaced it with a bisection-safeguarded Newton solve (a standard globally convergent technique, since $E - e\sin E - M$ is strictly monotonic and a containing bracket $[M-e, M+e]$ is known a priori), verified against 1.4×10^6 random (mean anomaly, eccentricity) pairs up to $e = 0.9999$ (worst-case residual 2×10^{-12}), and confirmed to leave every other result in this paper unchanged to within ordinary bootstrap resampling noise (Table 3's $\dot{\omega}$ precision shifts from 0.0013 to 0.0014 deg yr^{-1}, consistent with a small number of previously silently corrupted bootstrap resamples now computed correctly). With the underlying solver fixed, the Roemer-aware fit converges to the true parameters exactly (zero residual to machine precision) on noise-free data. That is a statement about bias, not about precision, so we tested precision separately, the same way frame tie's real bootstrap cost (Section 4.2) was found rather than assumed away: refitting a noisy, Roemer-corrected dataset with the self-consistent light-time-aware model and bootstrapping gives $\dot{\omega}$ precision of 0.0011–0.0012 deg yr^{-1}, statistically indistinguishable from Table 3's light-time-naive baseline. Unlike the frame tie, whose joint fit costs a real, measured $\sim$2.5$\times$ in precision (Section 4.2), the Roemer correction

Table 4. Comprehensive error budget for $\dot{\omega}$.

Term	Type	Effect (deg yr^{-1}, σ)	Status
Photon noise	Random	±0.0011 (1σ)	Baseline (Table 3)
Extended mass	Astrophysical	-0.0001 to -0.0003 (0.1–0.3σ)	Subdominant, bounded
Frame tie, unmitigated	Instrumental	-0.038 to -0.109 (33–94σ)	Dominant if ignored
Frame tie, joint fit	Instrumental	residual <0.3%; precision 0.0028	Fixed w/ real priors
Source confusion	Instrumental	< 1% precision change	Subdominant
Roemer delay, unmitigated	Geometric	-0.0017 (0.75%, 1.3σ)	Fails if ignored
Roemer delay, corrected	Geometric	residual 0 (exact)	Fixed w/ correct model
Sgr A* spin	Degeneracy	$+0.0066$ to $+0.0133$ (6.0–12.0σ)	Unbounded, out of scope

Note—Sections 4.1 (extended mass), 4.2 (frame tie), 4.3 (confusion), 4.4 (spin), and 4.5 (Roemer delay) give each term's real citation sources and methodology. Unmitigated Roemer delay also biases the orbital orientation angles i, Ω, ω by 14–16σ, not shown above; this is corrected along with $\dot{\omega}$ once the model accounts for light time.

here is genuinely free: no new free parameters, no bootstrap penalty, and no residual bias—fully corrected in precision as well as in bias, not just asserted to be from the noise-free check alone.

5. DISCUSSION

Comprehensive error budget. Table 4 collects every term quantified in Sections 4.1–4.5, alongside the random error from Table 3. Two things do not fit into a single "total uncertainty" number, and conflating them would misstate what this pipeline has actually shown:

First, the random and instrumental terms combine coherently once each is mitigated. The frame-tie joint fit already subsumes photon noise, giving 0.0028 deg yr^{-1}. Adding the extended-mass and confusion contributions in quadrature moves that figure by < 1% (e.g., $\sqrt{0.0028^2 + 0.0003^2} = 0.002816$). So $\sim$0.0028 deg yr^{-1} is the honest bottom-line *precision* of this pipeline, not the raw number quoted in Table 3; it is roughly 2.5$\times$ that frame-perfect figure. Roemer delay was once an uncorrected bias of comparable size. It is now fully removed once the model accounts for light time (§4.5). That correct treatment requires no ongoing quadrature term, the same way the frame tie needs none once it is fit jointly.

This also answers why the campaign does not simply request substantially more epochs. We tested a doubled-budget design (262 epochs, same periapsis-floor-plus-Fisher-optimal logic as Design D) and bootstrap-fit it: the random-error-only precision improves from 0.0011 to 0.00081 deg yr^{-1}, a real $\sim$26% gain, consistent with the $\sqrt{N}$ scaling more data should buy. But the pipeline's actual bottom line is not the random-error number; it is $\sim$0.0028 deg yr^{-1}, set mainly by the frame-tie systematic's own prior width (P. M. Plewa et al.'s real measured drift uncertainty), a fixed instrumental limit that does not shrink with more epochs the way random error does. Past the point where random error is already subdominant to a fixed systematic, doubling the telescope-time request buys a real but comparatively small improvement in the number that actually matters. This pipeline is already past that point.

Second, and most importantly, none of this bounds the spin systematic. It is 3–6$\times$ larger than the mitigated precision, and it has no analogous limit to inject. This is the fundamental reason the result is described as a *spin-agnostic* test throughout this paper, not a spin-free one: at real, plausible spin values, a measured $\dot{\omega}$ could differ from the pure-Schwarzschild prediction by more than everything else in Table 4 combined. The *Scope* paragraph below spells out concretely what "spin-agnostic" does and does not mean. Closing the spin gap itself is exactly T. Piran et al. (2026)'s multi-star program, not a numerical refinement of the single-star pipeline presented here.

Instrumentation: astrometry. No operating alternative beats GRAVITY+ for the astrometric measurement this paper proposes. The comparison comes down to raw resolution. Roman's 2.4 m aperture gives a diffraction limit of order 223 mas, larger than S301's entire 83 mas orbit; Roman cannot resolve the orbit's shape at all, let alone its precession. JWST (the James Webb Space Telescope)'s larger 6.5 m aperture is sharper ($\sim$85 mas), but JWST is still a single telescope, not an interferometer. Its resolution remains comparable to the orbit's own angular scale. GRAVITY+'s ∼130 m effective interferometric baseline reaches a much smaller fraction of that scale (∼100 μas, Section 3).

Instrumentation: radial velocity. No alternative helps here either, for three separate reasons. First, ERIS/SPIFFIER itself cannot deliver useful precision: Section 3 scales its real, achieved S2 precision to S301's fainter magnitude and predicts ∼1621 km s^{-1} against a ±15,000 km s^{-1} signal, only ∼9–13× better than the signal itself, and just 3.7% of SPIFFIER's own 60 km s^{-1} resolution element—nowhere near enough to pin down a line centroid. This is why RV was never proposed for the actual campaign in the first place; it is not a capability that was tried and then abandoned. Second, Roman fails independently of integration time. Its point spread function (PSF, the blurred shape a single point of light forms on the detector) is not adaptive-optics-corrected, so source confusion in this crowded field is a structural floor that does not average down with longer exposures. Roman's coarser spectral resolution ($R \approx$ 461–890 vs. ERIS's 5000) compounds the problem. Third, JWST/NIRSpec is more ambiguous. Its real achieved resolution is comparable to or better than pre-launch expectations (A. J. Shajib et al. 2025), and it has no ground-based OH airglow to contend with. But real observations of this specific field report severe, Galactic-Center-specific operational problems (detector saturation on bright cluster members, cross-talk between micro-shutters in the field, and misidentification of guide stars in the crowded core; F. Yusef-Zadeh et al. 2025). No published achieved precision exists yet for a point source as faint as S301. We therefore do not adopt JWST/NIRSpec as a demonstrated alternative.

Instrumentation: what HARMONI on the ELT could add. HARMONI, on the ELT (Extremely Large Telescope), deserves a closer look than a simple schedule dismissal. It is not just a bigger version of GRAVITY+: its 39 m aperture and K-band integral-field spectroscopy, at up to $R \approx$ 17,385 (N. Thatte & et al. 2024), address the RV problem specifically, not just astrometry. We extended this project's own validated RV scaling (Section 3: a real achieved SINFONI/ERIS result, scaled by flux in the same background-limited regime documented for this field) to HARMONI's much larger collecting area and finer spectral resolution. That extrapolation predicts $\sigma_{\rm RV} \approx 21$ km s^{-1} against S301's ±15,000 km s^{-1} signal, a signal-to-noise ratio (SNR) of order 10^3: genuinely feasible RV, not a marginal improvement. This prediction is a first-principles extrapolation to an instrument that has not yet observed anything; it is not a scaling of an achieved HARMONI result the way the ERIS number is. It should be read as motivation for follow-on work, not a proposal-ready sensitivity figure. On timeline, ESO's latest published schedule has HARMONI's science first light in December 2030 (ESO 2026), about ten months before passage 1 (October 2031). That is a tight window for a new instrument to clear proposal review, but not obviously an impossible one if preparatory work starts now and no further delay occurs. The honest statement: passage 1 is a stretch goal for HARMONI, while passage 2 (2040.5) is comfortably within reach. Adding a real RV channel by passage 2 is exactly the kind of advantage worth planning a proposal around, not writing off.

Scope. This paper's result is called a *spin-independent* (equivalently, *spin-agnostic*) test of general relativity, and that phrase needs a concrete definition. The quantity this pipeline measures, $\dot{\omega}$ in Equation 1, is the Schwarzschild (first-post-Newtonian, monopole-mass) apsidal precession. Any sufficiently compact central mass produces this precession, whether or not it spins, and regardless of how fast it spins if it does. It is the same effect that advances Mercury's perihelion around the Sun, a body whose spin plays no role in that calculation. Lense-Thirring precession is different: frame-dragging from Sgr A*'s own spin produces its own nodal precession, and, for a spin aligned with the orbit, its own additional apsidal precession too (Section 4.4). That spin-dependent signal was S301's original discovery motivation. *Spin-independent* means this measurement does not need to know or assume Sgr A*'s spin for the Schwarzschild test to be valid. It does *not* mean spin has zero effect on the specific number this pipeline reports. Section 4.4 already shows the opposite: if Sgr A*'s real spin is nonzero and aligned with S301's orbit, Lense-Thirring precession adds directly to the same $\dot{\omega}$ this pipeline fits, and at real published spin estimates this contamination reaches 6.0–12.0σ relative to the random-error precision (or 2.4–4.7σ relative to the wider, frame-tie-marginalized precision of Section 4.2; either way, this is comparable to or larger than every bounded systematic in Table 4). That is exactly why Table 4 lists spin as an open, unbounded caveat rather than a closed, bounded systematic like the other three terms. The test itself needs no spin measurement to be a valid probe of general relativity near Sgr A*; the specific fitted value could still be biased if Sgr A*'s true spin is nonzero, and that possibility is not resolved by this paper.

Measuring Sgr A*'s spin is explicitly outside this work. So, for now, is real data: like this paper, T. Piran et al. (2026) is a forecast, not a completed measurement. No periapsis passage has occurred yet for either analysis to use. Spin causes a distinct signature, nodal precession from frame-dragging, and this signature was not injected into this project's synthetic campaign. T. Piran et al. (2026) present the methodology a real spin measurement would additionally require: separating that spin signal from Newtonian confusion via multi-star differential calibration. That is a substantially harder analysis than the single-star, spin-independent measurement presented here. Concretely, this study's contribution to that eventual measurement is the part it actually completes: a validated light-curve and astrometric-fitting pipeline, a demonstrated observing cadence, and the systematic error budget in Table 4. That is the groundwork the harder, multi-star spin analysis would be built on top of, not a step of that analysis itself.

Design lesson. Local Fisher-information design assumes the model is well approximated by its linear expansion across the *entire* candidate range. That assumption fails badly for nonlinear periodic problems evaluated over multi-cycle baselines, and the failure stays invisible to the analytic bound itself.

A reasonable question is why this paper did not simply adopt Design D (Section 3.2) from the start, skipping Designs B and C entirely. The honest answer is that Design D's specific fix, a hard periapsis floor on top of Fisher-optimal allocation, is not obvious in advance. It only becomes obvious after seeing exactly how Designs B and C fail. Design B's failure (skipping periapsis to chase temporal-extreme clusters) revealed that pure Fisher optimization, evaluated at one guess, is not robust to being wrong about the true orbit. Design C fixed that specific weakness, by averaging Fisher information over many draws from a parameter prior instead of one fixed guess, a technique called pseudo-Bayesian prior-averaging (K. Chaloner & I. Verdinelli 1995). But Design C *still* skipped periapsis, which revealed a second, different weakness: Fisher information of any kind, prior-averaged or not, cannot detect geometric non-identifiability, the failure that comes from sampling too few distinct orbital phases regardless of how precisely each one is measured. Design D's periapsis floor directly targets that second, previously-hidden weakness. Skipping straight to Design D would have skipped the diagnostic work that identified which weakness needed fixing, and why.

We recommend three safeguards for any similar design problem: pseudo-Bayesian prior-averaging (Design C's fix for sensitivity to one fixed parameter guess), an explicit domain-knowledge floor against non-identifiability (Design D's fix for the failure prior-averaging cannot see), and validation against a real nonlinear fit rather than trust in the analytic prediction alone (without which neither failure would have been caught at all).

6. SUMMARY

A relativistic light-curve model reproduces S301's published solution to <0.1%. It predicts an asymmetric beaming feature, not an occultation dip: nothing physically blocks S301's light along our line of sight, since Sgr A* is a compact source, not an extended body capable of eclipsing it (Section 3.1). That feature is real but small (0.116 mag peak-to-trough) against S301's 0.30 mag photometric uncertainty; it is not visually detectable in this campaign's simulated photometry, and this paper's science result rests on astrometry, not photometry, for exactly that reason (Section 3.3). A feasibility calculation shows ERIS cannot detect S301's RV at any useful precision (Section 5). That is why this campaign proposes astrometry only, rather than including a channel known in advance to fail. A naive Fisher-optimal cadence catastrophically fails via nonlinear aliasing, despite an excellent analytic prediction. A hybrid design (a domain-knowledge floor plus a Fisher-optimal remainder) fixes this failure and beats a simple heuristic on every parameter. The resulting astrometry-only, two-passage GRAVITY+ campaign recovers the injected Schwarzschild precession rate (0.2307 deg yr^{-1}) as 0.2315 ± 0.0011 deg yr^{-1}. A non-circular check confirms this geometric result independently: it agrees with what the 1PN formula predicts from the same fit's own period and eccentricity (0.17σ), without either quantity referencing the other. A numerical post-Newtonian integration of S301's real orbit, using G. F. Rubilar & A. Eckart (2001)'s methodology and GRAVITY Collaboration et al. (2022)'s current extended-mass bound, shows that the leading astrophysical systematic, Newtonian confusion, biases $\dot{\omega}$ by only 0.12–0.29σ of the random measurement error, even at the conservative 3σ mass bound; that systematic is subdominant. The leading instrumental systematic is not subdominant: the astrometric reference-frame tie, at the real magnitude GRAVITY Collaboration et al. (2020) report as limiting their own S2 measurement, biases a naive fit's $\dot{\omega}$ by 17–47% of the entire signal (33–94σ) if left unmodeled. Fitting the frame tie jointly with real Gaussian priors, exactly as GRAVITY Collaboration et al. (2020) do, recovers $\dot{\omega}$ to within 0.3% of truth. The final precision ($\sim$0.0028 deg yr^{-1}) is only $\sim$2.5$\times$ wider than the frame-perfect case. A Monte Carlo bound on the remaining instrumental term, source confusion, shows that GRAVITY's real, quoted interferometric advantage (200–500$\times$ less affected than single-aperture imaging) keeps this term comparable to or below the photon-noise floor already modeled; source confusion changes $\dot{\omega}$'s precision by $< 1\%$. Two further checks probe the edges of this budget. First, an aligned Sgr A* spin, at real published estimates ($\chi = 0.9$–1), would contaminate $\dot{\omega}$ at the 6–12σ level relative to the random-error precision, with no analogous bound available to subtract it. That unresolved contamination is the genuine reason this result remains a spin-*independent* (spin-*agnostic*) test rather than a spin-*free* one, in the sense defined in Section 5's *Scope* paragraph: the test itself needs no spin measurement to be valid, but a real, nonzero spin could still bias the reported number. Second, a real Roemer (light-travel-time) delay, up to several days for this orbit, would bias $\dot{\omega}$ and distort the recovered orbital orientation by up to 16σ if ignored. A self-consistent light-time model fully corrects this delay, with zero residual bias. Making that correction converge robustly surfaced and repaired a genuine divergence bug in the project's core Kepler-equation solver; we verified the fix does not otherwise change any result in this paper beyond ordinary bootstrap noise. This is a synthetic forecast, not a completed measurement: real confirmation awaits the actual 2031.8 and 2040.5 passages. We conclude that, once observed, a two-passage GRAVITY+ campaign would be feasible with existing instrumentation. It would be sufficient for an independent, spin-agnostic GR test near Sgr A*, ahead of the harder spin measurement that S301's discovery motivated.

The Introduction states this paper's contribution briefly; here is the fuller version. This paper's contribution is best understood as two things, not one. The first is the feasibility case and error budget themselves. This is a numerically verified demonstration that existing instrumentation suffices, paired with a systematic-error treatment covering extended-mass confusion, the astrometric frame tie, source confusion, Roemer delay, and the spin degeneracy. That treatment is more complete on this specific question than either real S301 paper it builds on, in a concrete, comparable sense: GRAVITY Collaboration et al. (2026) forecasts spin sensitivity without a systematic error budget at all, so there is no instrumental-systematics table in that paper to compare against directly. T. Piran et al. (2026) addresses a harder, related problem: how to separate spin from Newtonian confusion across multiple stars once real data exist, not the error budget for a single star's monopole term. Neither omission is a gap in that prior work; their questions are simply different from this paper's. This paper's own systematic terms are not an independently invented noise model, either: each is anchored to a real, published instrumental figure from the same S-star literature, not an assumed number. The frame-tie magnitude comes from P. M. Plewa et al.'s own measured NACO/GRAVITY reference-frame drift. The confusion suppression factor (200–500×) is GRAVITY's own quoted interferometric advantage over single-aperture imaging. The extended-mass bound comes from GRAVITY Collaboration et al. (2022)'s current mass-distribution limit. The spin estimates come from GRAVITY Collaboration et al. (2026)'s own fiducial value and R. A. Daly & et al. (2024)'s independent outflow-method measurement. Section 4.1 already draws the closest direct numeric comparison available: this paper's disk/Lense-Thirring confusion ratio for S301 is qualitatively consistent with T. Piran et al. (2026)'s own finding that S301 is comparatively well-protected against that systematic, roughly 10× better than S2. It means the numbers in Table 4 did not exist anywhere in the literature before this analysis, for this star and this measurement, even though every input to them is drawn from published work.

The second contribution is methodological, and it emerged rather than being planned. The same failure pattern surfaced three independent times. A Fisher-information cadence design was excellent analytically and catastrophic in a real nonlinear fit (Section 3). An astrometric frame tie, when ignored, biased $\dot{\omega}$ by tens of percent despite the underlying data being fine (Section 4.2). A Kepler-equation solver diverged silently for ordinary parameter values that a fitting routine would actually visit; this bug was caught only because a robust fit was demanded of it (Section 4.5). In each case the fix was not cleverness. The fix was refusal to trust an analytic or first-pass result without a real nonlinear stress test. That pattern, not any single number in Table 4, is the more transferable lesson. For this class of problem (nonlinear, periodic, evaluated over a multi-cycle baseline, at the edge of an instrument's design regime), an untested analytic shortcut is not a conservative default. It is a specific, identifiable risk.

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This work uses the published orbital solution of GRAVITY Collaboration et al. (2026), and engages with T. Piran et al. (2026)'s multi-star Newtonian-confusion methodology for comparison in Section 4.1, without adopting it directly. Code, figures, and the precession animation are available at https://github.com/jcatanza/s301-schwarzschild-precession.

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